Jérome Eertmans

Ph.D. student in Differentiable Ray Tracing for Telecommunications

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Google Scholar

Education

Université catholique de Louvain, Ph.D. in Electrical engineering — Sept 2021 – June 2026
Thesis title: Differentiable Ray Tracing for Telecommunications

Université catholique de Louvain, M.Sc. in Electro-mechanical engineering — Sept 2019 – June 2021
Majored in mechatronics.

- I undertook a variety of course topics: telecommunications, open source software, numerical simulations, digital design of ICs, machine learning, mining pattern in data, etc., for a total of 112% workload.
- Graduated Magna Cum Laude.

Experience

Teaching Assistant, Université catholique de Louvain – Louvain-la-Neuve, Belgium — Sept 2021 – present

- Teaching duties (practical sessions, evaluation, …): see list here
- Research on my Ph.D. subject, under the supervision of Profs. Claude Oestges and Laurent Jacques.

Visiting Researcher, University of Bologna – Bologna, Italy — Sept 2024 – Dec 2024
Visiting Prof. Vittorio Degli-Esposti's lab and pursuing research on Machine Learning applied to Ray Tracing for Radio Propagation.

For additional experiences, see my LinkedIn profile.

Languages

French: Native
English: B2+/C1
Dutch: B1
Italian: A2
Japanese: Basics

Projects

DiffeRT

Python package with backend written in Rust.

- Built on top of JAX for high-performance and differentiable code
- Thoroughly tested and well documented
- Fully open access, along with its 2D variant DiffeRT2D

Manim Slides

Presentation tool I developed to produce higher-quality presentations at conferences.

- Most of my presentations are created using this tool and can be viewed on my personal website
- Used by other researchers to disseminate their research, with examples available in the documentation

More projects

You may find an extended list of my projects and contributions on my GitHub profile or on the projects page of my website.

Publications

Transform-Invariant Generative Ray Path Sampling for Efficient Radio Propagation Modeling Mar 2026
Jérome Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Nicola Di Cicco, Laurent Jacques, Claude Oestges
arxiv.org/abs/2603.01655 (arXiv preprint, submitted to npj Wireless Technology)

Fast, Differentiable, GPU-Accelerated Ray Tracing for Multiple Diffraction and Reflection Paths Oct 2025
Jérome Eertmans, Sophie Lequeu, Benoît Legat, Laurent Jacques, Claude Oestges
arxiv.org/abs/2410.14535 (arXiv preprint, accepted at EuCAP 2026)

Towards Generative Ray Path Sampling for Faster Point-to-Point Ray Tracing Oct 2024
Jérome Eertmans, Nicola Di Cicco, Claude Oestges, Laurent Jacques, Enrico Maria Vitucci, Vittorio Degli-Esposti
10.1109/ICMLCN64995.2025.11140249 (2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN))

Comparing Differentiable and Dynamic Ray Tracing - Introducing the Multipath Lifetime Map Oct 2024
Jérome Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Laurent Jacques, Claude Oestges
10.23919/EuCAP63536.2025.10999736 (2025 19th European Conference on Antennas and Propagation (EuCAP))

Fully Differentiable Ray Tracing via Discontinuity Smoothing for Radio Network Optimization Mar 2024
Jérome Eertmans, Laurent Jacques, Claude Oestges
10.23919/EuCAP60739.2024.10501570 (2024 18th European Conference on Antennas and Propagation (EuCAP))

DiffeRT2d - A Differentiable Ray Tracing Python Framework for Radio Propagation June 2024
Jérome Eertmans, Claude Oestges, Laurent Jacques
10.21105/joss.06915 (Journal of Open Source Software)

Min-Path-Tracing - A Diffraction Aware Alternative to Image Method in Ray Tracing Mar 2023
Jérome Eertmans, Claude Oestges, Laurent Jacques
10.23919/EuCAP57121.2023.10132934 (2023 17th European Conference on Antennas and Propagation (EuCAP))

Manim Slides - A Python package for presenting Manim content anywhere Aug 2023
Jérome Eertmans
10.21105/jose.00206 (Journal of Open Source Education)

Grants and Awards

Best Propagation Paper Award – Dublin, IE Apr 2026

EurAAP - EuCAP 2026.

TICRA-EurAAP Travel Grant – Dublin, IE Apr 2026

EuCAP 2026 travel grant.

Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (FNRS) – Brussels, BE Sept 2024 – Dec 2024

FRS FNRS *Accomplishment of a stay abroad*.

Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (WBI) – Brussels, BE Sept 2024 – Dec 2024

WBI *Bourses d'Excellence WORLD*.

STSM CA20120 - Visit to Pr. Vittorio Degli-Esposti – Brussels, BE Apr 2024

Cost Action Short-term Scientific Missions.

Participation in EuCAP 2024 – Brussels, BE Mar 2023

FRS FNRS *Participation in a scientific meeting*.

Personal Achievements

ADE Scheduler: An open source alternative to UCLouvain's scheduling tool, that became the official university tool in Sep. 2024.

Manim Slides: An open source presentation tool, used by many for teaching, conference presentations, or thesis defenses.

Scouts: 6 years as a scout (Les Scouts) leader.

Reviews

EuCAP: EuCAP 2026 (5 reviews)

IEEE: Access (5 reviews), Transactions on Vehicular Technology (6 reviews), Antennas and Wireless Propagation Letters (1 review), and Open Journal of the Communications Society (1 review)

Journal of Open Source Software: 1 review (see #9384)

Nature Partner Journals: Wireless Technology: 1 review

Referees

Prof. Claude Oestges, Université catholique de Louvain: claude.oestges@uclouvain.be

Prof. Laurent Jacques, Université catholique de Louvain: laurent.jacques@uclouvain.be

Prof. Vittorio Degli-Esposti, University of Bologna: v.degliesposti@unibo.it